

Team members

NETWORK DESIGN GROUP PROJECT

ROLES

- Project Manager- Chelsea Offei
- Network Topology - Jeremiah and Adam
- VLAN Planning - Conor
- Subnet Strategy-Conor
- Security Plan and Network Monitoring- SEIM - Chelsea
- Redundancy Plan -Gabriel
- Technical Writing - Adam
- Final Editing - Ben
- Presentation Design - Jada

Plan

Chelsea Offei

Company Type:

(Example: Office with HR, IT, Sales)

Network Size:

- Number of users: 60 employees
- Number of floors: 3 floors

Devices:

- Router(s): 2 Routers (1 primary router and 1 backup router)
- Switches: 2-core distribution switches (Only need 2)
- Access Points: 12 access switches (48 ports)(connect user devices)
- Wi-Fi: 3 wireless access points per floor
- Servers: 5 servers
- Firewall
- VoIP phone system

Explanations: The network will include routers, switches, wireless access points, servers, a firewall, and a VoIP phone system. The router connects the company network to the internet, while switches connect devices within the building so they can communicate. Wireless access points provide Wi-Fi for laptops and mobile devices. Servers store company data and run important network services. The firewall protects the network from unauthorized access and security threats. The VoIP system allows the company to make phone calls over the network. These devices work together to create a secure, reliable, and efficient network for the company.

VLANs:

- VLAN 10 → finance
- VLAN 20 → HR
- VLAN 30 → R&D
- VLAN 40 → Creative

Cabling:

- Ethernet (Cat6)- Used for connecting office computers, VoIP phones, printers, and wireless access points to access switches on each floor
- Fiber (if needed)- Only as the backbone between floors and for connections between core switches and routers.

Backup Plan:

- Backup router: One backup router will be installed in case the primary router fails.
- Redundant core switches: Core switches will be set up so that if one switch fails, traffic can still pass through another switch.
- Secondary internet connection: A second internet connection will be available so the company can stay online if the main connection goes down.
- Server backups: Important company data and system files will be backed up regularly to prevent data loss.

Failover idea :

- Failover support: If one device or connection fails, traffic will automatically switch to the backup path or backup device

Network Topology

Network Topology

Name: Jeremiah Thao & Adam Massey

- Create diagram:
 - Topology Diagram by Gabriel: Need to add VLANs
 - https://lucid.app/lucidchart/c54aa6b7-fe8c-481e-9005-a9b7e5db1bc5/edit?viewport_loc=-2895%2C536%2C4009%2C1816%2C0_0&invitationId=inv_3dcbb88d-9117-4901-b42e-687d483748bc
- Follow the Master Plan Diagram and explanation

Discuss your plans for the physical infrastructure and the topology of the network. You should discuss your plans for the following :

- **The Blueprint:** A professional-grade topology diagram. You must show the **Core, Distribution, and Access layers**.
- **Media Selection:** Explicitly state where you are using Cat6a vs. Fiber Optic (e.g., 10GbE Fiber for the backbone, Cat6a for the workstations).
- **Wireless Density:** A "Heat Map" or description of Wi-Fi 7 Access Point placement to ensure the 40 laptops have seamless roaming and zero dead zones. You do not need to create a detailed blueprint, but you do need to discuss the range of the access points and how you will prevent zones with low access. Mention your plan to conduct a site survey, and what you would be looking for there.

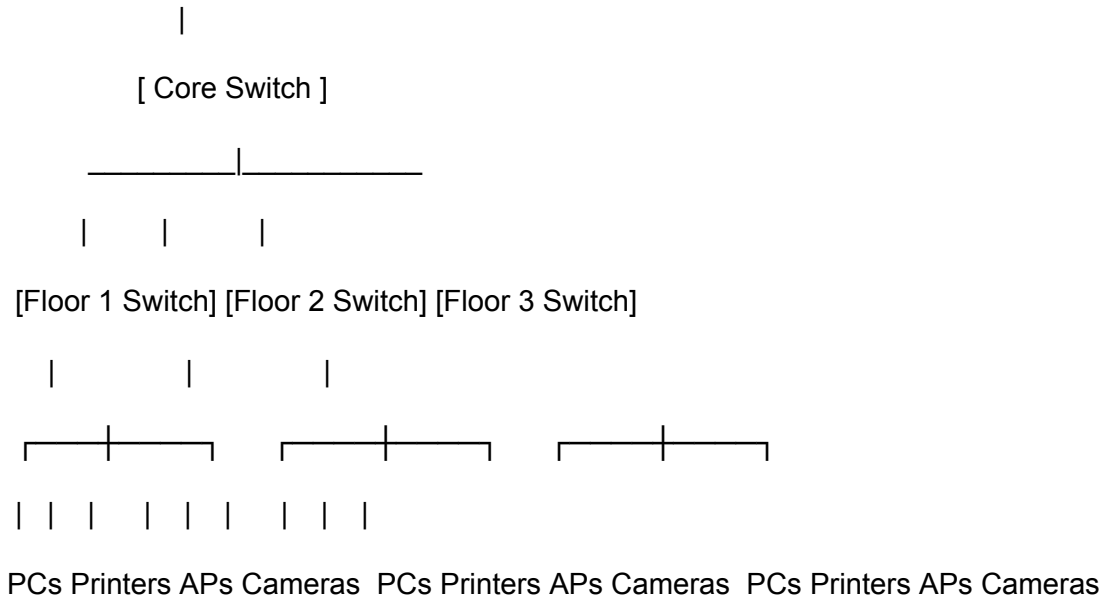
[Internet]

|

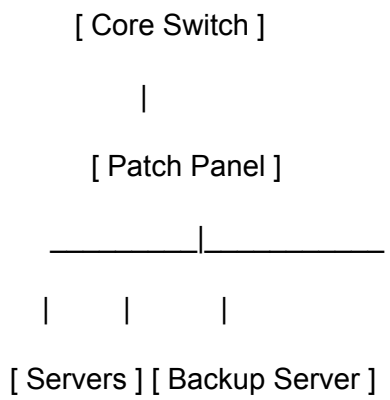
[Firewall]

|

[Router]

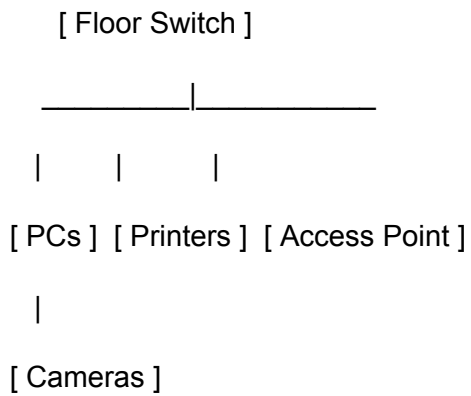


Server room diagram



|
[UPS]

Floor layout

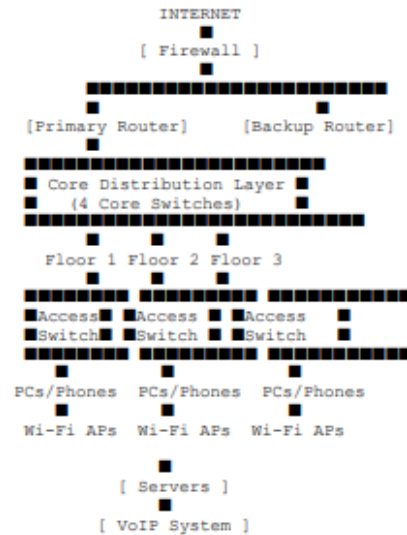


- The Internet connects to a firewall for security.
- The firewall connects to a router to manage traffic.
- The router connects to a core switch, which acts as the backbone.
- The core switch connects to switches on each floor.
- Each floor switch connects to PCs, printers, access points, and cameras.
- The server room contains servers, a backup server, a patch panel, and a UPS.

Network Topology Design

This document outlines a hierarchical network topology for a company with 60 employees across 3 floors. The design includes routers, switches, wireless access points, servers, a firewall, and a VoIP system.

Topology Diagram:



Layer Descriptions:

Edge Layer: Internet connects to Firewall, then to Primary and Backup Routers for redundancy.

Core Layer: 4 core switches act as the backbone connecting all floors and servers.

Access Layer: Each floor has access switches connecting user devices and wireless APs.

Wireless Network: 8 access points distributed across floors provide Wi-Fi coverage.

Server Segment: 5 servers provide services like storage, authentication, and applications.

VoIP System: Provides IP-based telephony connected through the network.

VLAN Strategy

VLAN Planning

Name: Conor Mayhorn

VLAN ID	VLAN Name	Purpose
10	Finance	Finance desktops and finance users
20	HR	HR desktops and HR users
30	R&D	R&D desktops and R&D users
40	Creative	Creative desktops and creative users
50	Corporate Wireless	Company laptops on Wi-Fi
60	Voice	VoIP phones

70	Printers	Network printers
80	Server Farm	Internal file servers and internal management systems
90	Management	IT admin access to switches, firewalls, APs, servers
100	DMZ	Public-facing web, mail, FTP services
110	Building IoT Wired	Climate control, wired badge/biometric systems, wired cameras
120	Building IoT Wireless	Smart lighting and wireless cameras
130	Remote Access VPN	VPN clients after authentication
140	Guest Wi-Fi	Optional guest or contractor internet-only access

999	Native/Blackhole/Unused	Unused ports and parking VLAN
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VLAN Breakdown:

Our VLAN strategy is designed to separate the Nexus Innovation Hub network by both department and device type. This improves security, reduces unnecessary broadcast traffic, and makes the network easier to manage. Since the company has sensitive departments like Finance and HR, as well as high-risk systems like IoT devices and public-facing servers, it is important to isolate different parts of the network rather than put everything on a single large shared network.

VLAN 10 – Finance

This VLAN is for the Finance department desktops and users. Finance handles sensitive financial records, so this VLAN should have stricter security policies and limited access from other departments.

VLAN 20 – HR

This VLAN is for Human Resources systems and users. HR data, including employee records and other private information, should be separated from the rest of the company network.

VLAN 30 – R&D

This VLAN is for the Research and Development department. Since R&D may work with valuable intellectual property and internal projects, this VLAN should be protected from unnecessary access by other departments.

VLAN 40 – Creative

This VLAN is for the Creative department. It separates creative workstations from other departments while still allowing access to approved shared resources such as file storage and printers.

VLAN 50 – Corporate Wireless

This VLAN is for the 40 company-issued laptops using Wi-Fi 7. Keeping wireless users on their own VLAN allows easier roaming, wireless security management, and better control over laptop access to internal resources.

VLAN 60 – Voice

This VLAN is for the 20 VoIP phones. A separate voice VLAN helps organize phone traffic and supports Quality of Service, which improves call quality and reliability.

VLAN 70 – Printers

This VLAN is for the 6 network printers. Printers are shared resources and are often security weak points, so placing them in their own VLAN helps control which users can reach them.

VLAN 80 – Internal Servers

This VLAN is for the internal server farm, including the 4 file servers and the dedicated SIEM/management server. These systems are critical to business operations, so they should be isolated from normal user traffic and only accessible through approved rules.

VLAN 90 – Management

This VLAN is for IT administrators and network device management. It would be used to manage switches, routers, firewalls, and other infrastructure. Only authorized admins should have access to this VLAN.

VLAN 100 – DMZ

This VLAN is for public-facing services such as the web, mail, and FTP servers. The DMZ creates a buffer zone between the internet and the internal network so that if one public system is compromised, attackers cannot easily reach internal department VLANs.

VLAN 110 – Wired IoT

This VLAN is for wired IoT devices such as climate control systems, biometric/swipe access systems, and wired security cameras. These devices should be isolated because IoT equipment often has weaker security than normal business systems.

VLAN 120 – Wireless IoT

This VLAN is for wireless IoT devices such as smart lighting and wireless cameras. Separating wireless IoT from the main business network reduces risk and supports the project's requirement for strong IoT isolation.

VLAN 130 – Remote Access VPN

This VLAN is for remote employees after they connect through the company VPN. This allows remote users to be authenticated first and then given only the access they need based on least privilege.

Explanation:

Overall, this VLAN design supports security, efficiency, and scalability. Departments are separated to protect sensitive data, infrastructure devices are grouped by function for easier management, and risky systems like IoT and public-facing servers are isolated from critical internal resources. This structure also makes it easier to apply ACLs and firewall rules to control traffic between VLANs.

Subnet Strategy

Host requirements

VLAN	Name	Estimated Hosts
10	Finance	30
20	HR	30
30	R&D	30
40	Creative	30
50	Wireless	40
60	Voice	20
70	Printers	6
80	Server Farm	5
90	Management	15
100	DMZ	3-6
110	IoT Wired	110
120	IoT Wireless	100
130	VPN	30
140	Guest WI-FI	20

Final subnetting table

VLAN	Name	Prefix	Subnet	Usable Range	Broadcast
110	IoT Wired	/25	10.20.0.0	.1 – .126	.127
120	IoT Wireless	/25	10.20.0.128	.129 – .254	.255

10	Finance	/26	10.20.1.0	.1 – .62	.63
20	HR	/26	10.20.1.64	.65 – .126	.127
30	R&D	/26	10.20.1.128	.129 – .190	.191
40	Creative	/26	10.20.1.192	.193 – .254	.255
50	Wireless	/26	10.20.2.0	.1 – .62	.63
130	VPN	/26	10.20.2.64	.65 – .126	.127
60	Voice	/27	10.20.2.128	.129 – .158	.159
90	Managemen t	/27	10.20.2.160	.161 – .190	.191
70	Printers	/28	10.20.2.192	.193 – .206	.207
80	Servers	/28	10.20.2.208	.209 – .222	.223
100	DMZ	/28	10.20.2.224	.225 – .238	.239
140	Guest Wi-Fi	/27	10.20.3.0	.1 – .30	.31

Default Gateway Plan

VLAN	Gateway
Finance	10.20.1.1
HR	10.20.1.65
R&D	10.20.1.129
Creative	10.20.1.193

Wireless	10.20.2.1
IoT Wired	10.20.0.1
IoT Wireless	10.20.0.12

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DHCP Integration

VLAN	DHCP Range
Finance	10.20.1.10 – 10.20.1.60
HR	10.20.1.70 – 10.20.1.120
Wireless	10.20.2.10 – 10.20.2.60
IoT Wired	10.20.0.10 – 10.20.0.120

Subnet Calculations

<https://drive.google.com/file/d/14ggYKlPF5bjB61PlrcIFdI7Psm2coHwG/view?usp=sharing>

Security Plan

Security Plan - Chelsea

1. Firewall Security

A firewall will be placed between the Internet and the internal network to monitor and control incoming and outgoing traffic. The firewall will block unauthorized access and only allow approved traffic into the network. The firewall will also protect the internal network from external cyber threats.

2. VLAN Security

The network is divided into VLANs to improve security and reduce unnecessary traffic between departments. Sensitive departments such as Finance and HR are placed in separate VLANs to protect confidential data. IoT devices, guest Wi-Fi, and public servers are also placed in separate VLANs to prevent unauthorized access to the internal network.

3. Access Control Lists (ACLs)

ACLs will be configured to control traffic between VLANs. For example:

- Finance VLAN can access the servers, but not the HR VLAN
- HR VLAN can access the servers, but not the Finance VLAN
- Guest Wi-Fi can only access the Internet
- The IoT VLAN cannot access the internal department VLANs
- The management VLAN can access all network devices

This improves network security and follows the principle of least privilege.

4. DMZ (Demilitarized Zone)

Public-facing servers such as web, email, and FTP servers will be placed in the DMZ VLAN. The DMZ acts as a buffer zone between the Internet and the internal network. If a public server is attacked, the attacker will not have direct access to the internal network.

5. VPN Security

Remote workers will connect to the network using a secure VPN connection. After authentication, users will be placed into the Remote Access VLAN and will only be allowed to access the resources they are authorized to use.

6. Wireless Security

The wireless network will use WPA3 encryption to protect wireless communication. The guest Wi-Fi network will be separated from the internal network and will only allow internet access.

7. IoT Security

IoT devices such as smart lighting, cameras, and climate control systems will be placed in separate VLANs because IoT devices are more vulnerable to attacks. These devices will not be allowed to communicate directly with sensitive department VLANs.

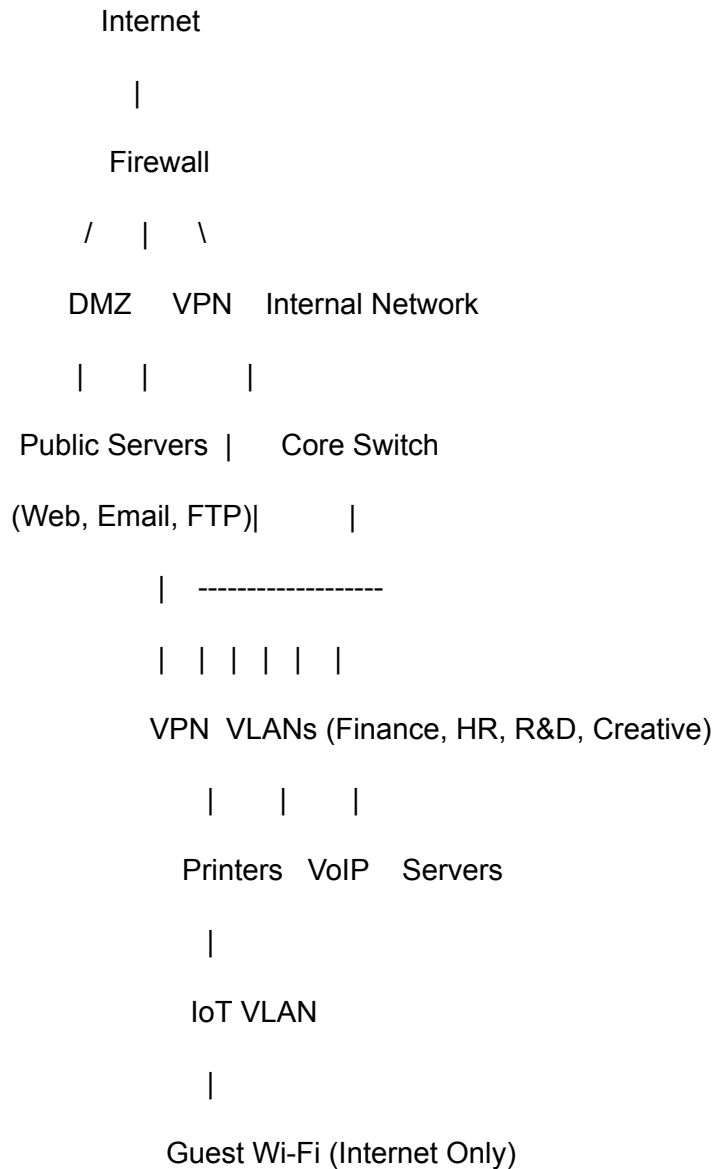
8. Port Security

Port security will be enabled on access switches to prevent unauthorized devices from connecting to the network. Only approved devices will be allowed to connect to switch ports.

9. Network Monitoring and SIEM

A SIEM (Security Information and Event Management) server will be used to monitor network traffic and detect suspicious activity. The SIEM system will alert administrators if a potential security threat is detected.

Security Diagram



- The SIEM Server monitors all traffic
- ACLs control traffic between VLANs

The SIEM system will monitor traffic across all VLANs, servers, and network devices and will generate alerts if suspicious activity or network failures are detected.

Network Monitoring

Network Monitoring

The network will be monitored using a Security Information and Event Management (SIEM) system and network monitoring tools. The monitoring system will track network traffic, device status, and network performance. If a device fails, unusual activity is detected, or a security threat occurs, the system will automatically alert the network administrator. Network monitoring improves network security, reliability, and performance by allowing problems to be detected and resolved quickly.

Monitoring Tool	Purpose
SIEM	Detect security threats
Network monitoring software	Monitor devices and traffic
Alerts	Notify admin if failure occurs
Logs	Record network activity
Performance monitoring	Track network speed and usage

Redundancy Plan

Redundancy Plan

Name: Gabriel Linares

- Plan a backup system

Details: **Backup plan**

- **Power backup**
- **Failover plan**
- **Disaster recovery**

Explanation:

Research: <https://www.auvik.com/franklyit/blog/simple-network-redundancy/>

Problem	Solution
Power outage	UPS
Server failure	Backup server
Data loss	Cloud backup
Switch failure	Spare switch
Internet failure	Secondary ISP (optional)

Redundancy Network Diagram :

“Draw a redundancy network diagram showing a main server and backup server connected to a core switch with a UPS battery backup system.”

Redundancy Planning:

- Dual Internet Service Providers
 - 1 external firewall needed for each ISP
 - 1 edge router needed for each ISP (running BGP for failover)
 - 2 core routers to provide redundancy
 - Routers running ECMP (High-availability)
 - Provides Failover
- Internal Network (after external firewalls)
 - 2 layer-3 switches for inter VLAN communication (VRRP, redundancy pair)
 - 12 layer-2 switches (4 per floor; accounting for company growth)
 - Each switch trunks all VLANs (for redundancy, HA)
 - VLAN pruning per floor
 - 48 ports each
 - Floor 1:
 - Floor 2:
 - Floor 3:
 - 2 Layer-2 switches for Internal File servers and SIEM
 - UPS units as necessary dictated by power load
 - 2 Layer-2 switches for DMZ
 - Wireless AP
 - At least 3 per floor
 - 1500-2000 sqft coverage each
 - Site survey needed
- Backup Power
 - UPS units
 - 1-2 per server rack
 - Depends on wattage consumed per server
- Topology Diagram
 - https://lucid.app/lucidchart/c54aa6b7-fe8c-481e-9005-a9b7e5db1bc5/edit?viewport_loc=-2895%2C536%2C4009%2C1816%2C0_0&invitationId=inv_3dcbb88d-9117-4901-b42e-687d483748bc
 -

Disaster Recovery Plan

Disaster Recovery Plan - Chelsea

The Disaster Recovery Plan for _____ ensures that the company can recover its network and data in the event of a major failure such as a power outage, hardware failure, cyberattack, or natural disaster. The goal is to minimize downtime and ensure business continuity.

- **Server Failure**

If the main server fails, the backup server located in the server room will take over operations. Important files and systems are backed up regularly so data can be restored quickly.

- **Power Outage**

If a power outage occurs, the Uninterruptible Power Supply (UPS) will provide backup power to critical devices in the server room, such as the core switch, servers, firewall, and router. This will allow the network to continue running for a short time or shut down safely.

- **Internet Failure**

If the primary ISP fails, the secondary internet connection will automatically take over to maintain internet connectivity for the company.

- **Network Device Failure**

If a router or core switch fails, the backup router and redundant core switch will take over network operations using failover protocols.

- **Data Loss**

All important company data will be backed up to both a local backup server and secure cloud storage. If data is lost due to hardware failure or cyberattack, the data will be restored from the backup systems.

- **Cyberattack**

If a cyberattack occurs, the firewall and security monitoring system will detect the threat and block unauthorized access. Systems can be restored using backup data if necessary.

Disaster Recovery Objective

The objective of the disaster recovery plan is to restore critical network services such as servers, internet access, and communication systems as quickly as possible to reduce downtime and maintain business operations.

Here is a table that makes explanations easy to understand.

Disaster Recovery Summary Table

Disaster	Recovery Solution
Power outage	UPS backup power
Server failure	Backup server
Internet failure	Secondary ISP
Router failure	Backup router
Switch failure	Redundant core switch
Data loss	Local + Cloud backup
Cyberattack	Firewall + Restore from backup
Natural disaster	Cloud backup + Restore system

Network Automation

Technical Writing

2Technical Writing

Name: Adam Massey

- Write all sections clearly
- You need to write explanations for:

- **Device**

A device is any physical piece of hardware that connects to a network to send, receive, or process data. Examples include computers, smartphones, printers, and IoT devices. Each device typically has a unique IP address that allows it to communicate with other devices on the network.

- **Router**

A router is a networking device that directs data traffic between different networks, such as a home network and the internet. It assigns IP addresses to connected devices and ensures data packets are sent to the correct destination. Routers also often include basic security features like network address translation (NAT) and firewalls.

- **Firewall**

A firewall is a security system that monitors and controls incoming and outgoing network traffic based on predetermined rules. It acts as a barrier between a trusted internal network and untrusted external networks, such as the internet. Firewalls help prevent unauthorized access and protect against cyber threats.

- **Core Switch**

A core switch is a high-capacity network switch that operates at the backbone of a network, connecting multiple distribution or access switches. It is designed for fast data transfer and efficiently handles large amounts of traffic. Core switches are typically used in large networks like businesses or campuses.

- **Access Switch**

An access switch connects end devices, such as computers and printers, to the network. It operates at the edge of the network and enables devices to communicate with one another and access network resources. Access switches are commonly used in offices, schools, and homes.

- **Access Point**

An access point (AP) is a device that provides wireless connectivity to a network. It allows Wi-Fi-enabled devices, such as laptops and smartphones, to connect to a wired network. Access points extend network coverage and improve wireless signal strength in larger areas.

- **Server**

A server is a powerful computer that provides services, resources, or data to other devices on a network. Common types include file servers, web servers, and application servers. Servers are designed to handle multiple requests simultaneously and operate continuously.

- **Backup Server**

A backup server is a system used to store copies of data from primary servers or devices. It ensures that important information can be recovered in case of data loss, hardware failure, or cyberattacks. Backup servers are essential for maintaining data integrity and business continuity.

- **Patch Panel**

A patch panel is a hardware component used to organize and manage network cables. It provides a central location where cables can be connected and easily reconfigured. Patch panels improve cable management and make troubleshooting and maintenance more efficient.

- **UPS (Uninterruptible Power Supply)**

A UPS is a device that provides backup power to network equipment during electrical outages. It helps prevent data loss and hardware damage by allowing devices to shut down properly. UPS systems also protect against power surges and voltage fluctuations.

- **Cat6 Cable**

A Cat6 cable is a type of Ethernet cable used for high-speed network connections. It supports faster data transmission and reduced interference compared to older cable types like Cat5. Cat6 cables are commonly used in modern networks for reliable and efficient communication.

Presentation

Presentation

Name: Jada Sapp

- Create slides
- Keep it simple

Team 9 - Network Consulting Proposal Project

Link below is the template being used for this presentation. If you're unable to view, please let me know.

Template

Copy of Software Engineering Business Plan by Slidesgo

(Ps) google slide

Sources

Don't Forget to Cite Your Sources! Put a reference slide at the end. You are not required to do full in-text citations. If you use an LLM to find information, cite it! Also, make sure you check any information from an LLM for accuracy; remember that LLMs can hallucinate. IEEE reference formatting is preferred, but MLA or APA is acceptable.

Using MLA 8th

Sources

Include sources used below this so I can transfer them to MLA 8th and have them for the reference slide, thanks!

NOTE: Any links in red indicate that they're transferred into the citations I plan to put onto the slides, please put any new links below the red, thanks.

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[Fortinet FortiGate 2026 Verified Reviews, Pros & Cons](#)

[Fortinet FortiGate: Pros and Cons 2026](#)

[Fortinet Firewalls Reviews 2026: Details, Pricing, & Features | G2](#)

<https://aws.amazon.com/marketplace/reviews/reviews-list/B00PCZSWDA>

[FortiGate Next-Generation Firewall Reviews, Pros and Cons - 2026 Software Advice](#)

[Fortinet Reviews, Ratings & Features 2026 | Gartner Peer Insights](#)

[Next Generation Firewall \(NGFW\) - See Top Products](#)

[Fortinet FortiGate 100F Series | AVFirewalls.com](#)

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Works Cited

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April 2026.

Pitches

Slide 1 - Title

We are Team 9 and this is our Networking consulting Proposal

Slide 2 - Team

Here are the members of our teams and their roles in the project

Slide 3 - Network Design Overview

This slide provides an overview of our network design.

Our goal was to create a network that is secure, dependable , and Flexible .

To achieve this, we used the FortiGate 100F for security and traffic control, VLANs to organize the network , improve performance, and backup systems to ensure the system stays running even if a failure occurs.

Together, these components create a network that is efficient and protected.

Next, we'll take a closer look at the network topology.

Conor Mayhorn Slides: 9-13

Now I'm going to walk through our logical network design, focusing on our VLAN strategy and subnetting plan, which are essential for organizing, securing, and managing the network. As shown in our slides, our approach was to segment the network in a way that supports both business operations and strong security practices.

Our VLAN strategy is based on separating the network by both departments and device types. In total, we created 14 VLANs to isolate traffic, reduce congestion, and enforce security policies. We first implemented department-based VLANs for Finance, HR, R&D, and Creative. These VLANs help isolate sensitive information. For example, Finance and HR handle confidential data, so keeping them separate reduces the risk of unauthorized access and limits the impact of potential threats.

In addition to department VLANs, we created VLANs for different types of devices. These include a Wireless VLAN for company laptops, a Voice VLAN for VoIP phones, a Printer VLAN, a Server VLAN, and a Management VLAN for administrators. We also included security-focused VLANs, such as a DMZ for public-facing services, separate VLANs for wired and wireless IoT devices, a VPN VLAN for remote users, and a Guest VLAN for visitors. This level of segmentation reduces broadcast domains, improves performance, and prevents unnecessary communication between devices. It also limits lateral movement, meaning that if one part of the network is compromised, it does not easily spread to other areas.

To support this VLAN design, we implemented a subnetting plan using Variable Length Subnet Masking, or VLSM. We selected the base network 10.20.0.0 slash 22, which provides over one thousand usable IP addresses and allows room for future growth. Instead of using one fixed subnet size, we assigned subnet sizes based on the number of devices in each VLAN to ensure efficient use of IP space.

For larger networks, such as the IoT VLANs, we used /25 subnets, which provide 126 usable IP addresses. Medium-sized networks, such as departmental VLANs, wireless users, and VPN users, were assigned /26 subnets, which provide 62 usable IP addresses. Smaller networks, such as printers, servers, and the DMZ, were assigned /27 or /28 subnets, depending on their size. This approach ensures that we are not wasting IP space while still allowing for scalability.

We determined these subnet sizes using a standard calculation. We take the number of required devices, add two to account for the network and broadcast addresses, and then choose the next power of two. For example, the Wireless VLAN supports 40 devices. Adding two gives us 42 required addresses, and the next power of two is 64, so we assign a /26 subnet. Similarly, for the Wired IoT VLAN with around 110 devices, we calculate 112 required addresses, which results in a /25 subnet. This ensures each VLAN has enough IP addresses without over-allocating resources.

Finally, our VLAN and subnet designs are fully integrated. Each VLAN is mapped to its own subnet, and default gateways are configured to allow communication between networks. In our design, inter-VLAN routing and security enforcement are handled by the FortiGate 100F, which acts as both a firewall and routing device. This allows all traffic between VLANs to be inspected and controlled using firewall policies.

We also use a centralized DHCP server to assign IP addresses dynamically, with DHCP relay configured to allow communication across VLANs. In addition, we apply Access Control Lists and firewall rules to control how traffic flows between VLANs. For example, IoT devices are restricted from accessing internal business systems, the DMZ is isolated from internal VLANs, and sensitive departments like Finance and HR have limited access from other parts of the network.

Overall, our VLAN and subnetting design creates a network that is secure, scalable, and efficiently organized, while the FortiGate 100F ensures that all traffic is properly routed, inspected, and protected.

Gabriel Linares Slides: 15-17

3.3 Core Services and the DMZ

I'll start by discussing our core network services, including DNS, DHCP, the SIEM system, and how we securely integrate them using a DMZ.

As my colleague Connor already mentioned for DNS and DHCP, we use a centrally managed architecture. This allows us to maintain a simplified administration plan, since your company has limited IT personnel. DHCP is responsible for dynamically assigning IP addresses within each VLAN using our predefined VLAN scopes, which ensures efficient IP management and prevents conflicts.

For DNS services, we must take into account internal and external resolution. For internal resources our internal DNS server is able to associate the local records to match an internal IP address. On the other hand, an external DNS server is responsible for resolving public services such as our website and email servers. This separation ensures that internal systems are never directly exposed to the public internet.

Next, we implement a DMZ, or demilitarized zone, to securely host public-facing services. This includes our web server, email server, FTP server. The DMZ sits between the external network and the internal network, acting as a buffer zone.

The key security component here is our firewall configuration. By default, we deny all traffic from the DMZ into the internal network. This is critical because it prevents attackers from moving laterally if a public-facing server is compromised.

For example, we explicitly block any traffic from the DMZ to the Finance VLAN. So even if a web server is hacked, the attacker cannot access sensitive financial systems. At the same time, we allow only specific, controlled traffic—such as HTTP and HTTPS requests from the internet to the web server, and limited internal communication for updates.

Finally, we integrate all core systems into our centralized Security Information and Event Management solution. The SIEM collects logs from multiple sources, including firewalls, switches, routers, servers, and even DNS and DHCP systems.

These logs are analyzed using detection rules to identify suspicious behavior. For example, the SIEM can detect an “impossible travel” scenario, where a user logs in from New York and then appears in California just minutes later. It can also detect unusual activity like an IoT device attempting to initiate an SSH connection to a file server, or excessive DNS queries. All of these

events are recognized by AI driven rules which are flagged in real time, allowing quick analysis and resolution of security threats.

3.4 Security

Now I'll move on to network security, which focuses on keeping the network secure even during failures or attacks.

We incorporate a Zero Trust remote access model for our remote employees. Instead of trusting users by default, every access request is verified.

The authentication process begins when a user attempts to connect. The request is intercepted by FortiGate firewalls, where a device posture check is performed to ensure the device meets security requirements. The user then authenticates using single sign-on and must complete multi-factor authentication. Only after these steps are completed is access granted, and even then, the user is given only the minimum permissions necessary. This prevents unauthorized access and limits the impact of compromised credentials.

Finally, we enforce strict IoT isolation to protect sensitive systems. IoT devices are placed into dedicated VLANs—one for wired devices and one for wireless devices.

Firewall rules create a “hard wall” between these IoT networks and the rest of the system. For example, if a smart thermostat is compromised and attempts to access the Finance VLAN, that request is immediately denied.

This containment strategy ensures that even vulnerable devices cannot impact critical business systems.

3.4 Business Continuity

Next, we address high availability. Our goal is to eliminate single points of failure across the network.

At the firewall layer, we use a pair of Fortigate 100F devices configured in an active-active cluster. This means both firewalls are actively handling traffic, which provides load balancing characteristics. Thus, preventing the network from being congested during high traffic events. If one fails, the other continues routing without interrupting network connectivity.

At the switching layer, we use LACP for combining multiple physical links into one logical connection for redundancy and RSTP for preventing network loops. This allows us to provide resiliency for internal services and a standby network path in case of failures.

This ensures that both external and internal communication remain uninterrupted during hardware failures.

Next, we address power continuity. All critical systems, including servers, firewalls, and switches, are equipped with dual power supplies and connected to backup power sources.

If one power supply fails, the system continues operating normally. In the event of a full power outage, the backup system provides enough time for a graceful shutdown. Thus, mitigating data corruption and protecting system integrity.

For data protection, we implement a hybrid backup strategy based on recovery objectives. We perform local backups hourly, which allow for quick recovery in case of minor failures. In addition, we perform weekly cloud backups to protect against large-scale disasters such as hardware failure, ransomware attacks, or site loss.

This ensures that data can always be restored with minimal loss and downtime.